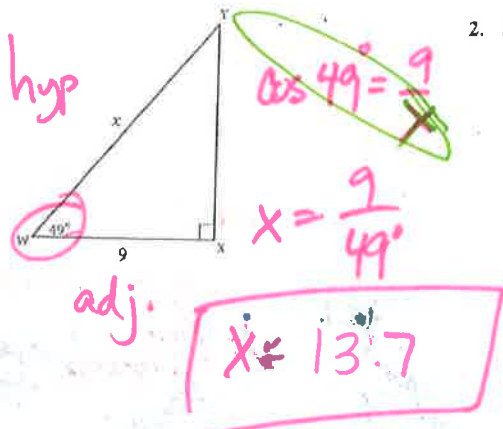


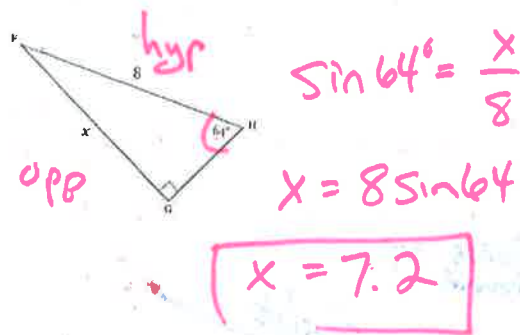
A2 Notes T.2 Inverse Trig and Word Problems

Warm Up

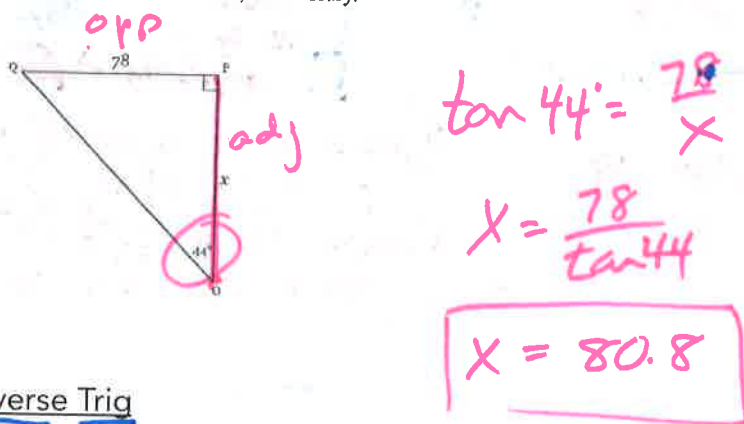
1. Solve for x . Round to the nearest tenth, if necessary.



2. Solve for x . Round to the nearest tenth, if necessary.



3. Solve for x . Round to the nearest tenth, if necessary.



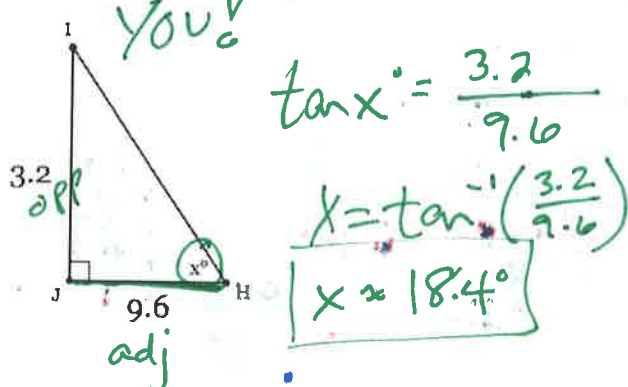
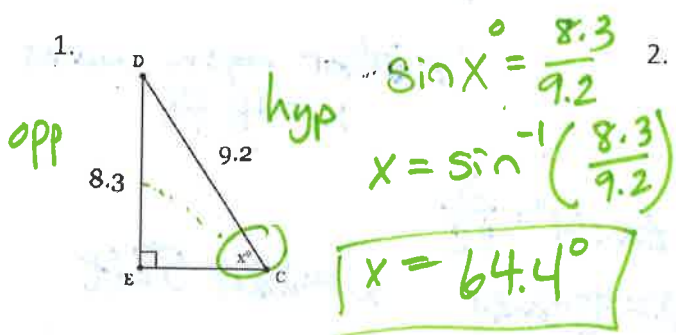
Notes on Inverse Trig

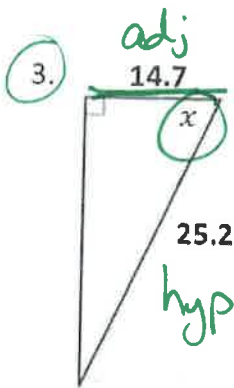
Last time, we had a side and an angle, and used SOH CAH TOA to find a missing side. Today, we will be given 2 sides, and will again use SOH CAH TOA, but this time to find the angle!

To do this, we use the INVERSE functions:

$$\sin^{-1} \neq \frac{1}{\sin} \quad \text{read "sign inverse"} \quad \sin^{-1} \left(\frac{\text{ratio of sides}}{\text{of}} \right) = \text{angle}$$

Examples. Give the figure, solve for x to the nearest tenth of a degree.

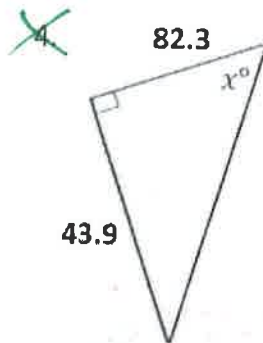




$$\cos x^\circ = \frac{14.7}{25.2}$$

$$x = \cos^{-1}\left(\frac{14.7}{25.2}\right)$$

$$x = 54.3^\circ$$



Notes Angles of Elevation and Depression

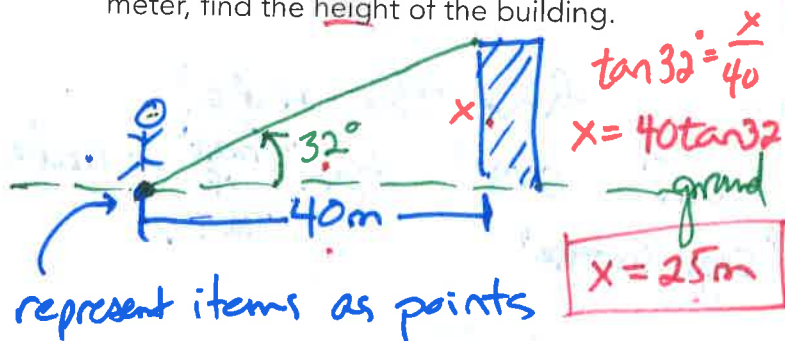
ant looks up
from ground



measure FROM Horizontal up
to line of sight

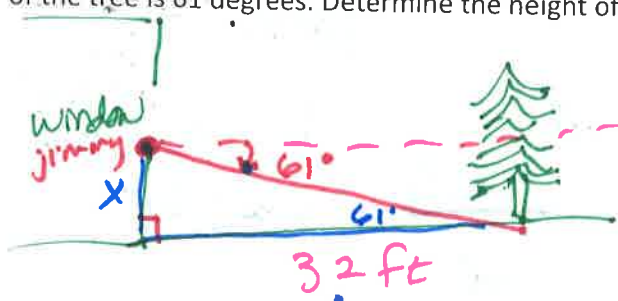
Examples.

1. Mateo stands 40 m from a building. The angle of elevation from his foot to the top of the building is 32° . To the meter, find the height of the building.



represent items as points

3. Jimmy is looking out his window at a tree that is 32 ft from his house. The angle of depression to the bottom of the tree is 61 degrees. Determine the height of the window.



$$\tan 61^\circ = \frac{x}{32}$$

$$x = 32 \tan 61^\circ$$

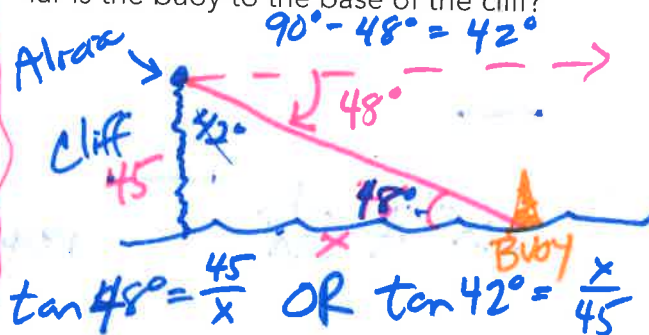
$$x = 58\text{ ft}$$

gazes along horizontal



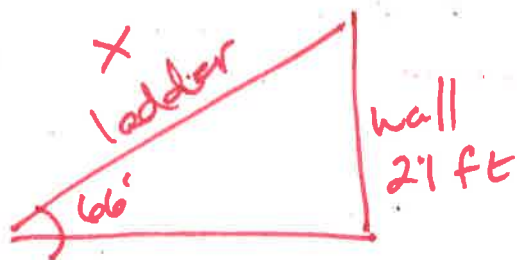
Note! $\angle$ of depression is OUTSIDE OF OUR Rt. Δ .
Measure from HORIZONTAL down to line of sight

2. Alicia observes a buoy in the sea below her from a 45 ft cliff. The angle of depression from Alicia's foot to the buoy is 48° . How far is the buoy to the base of the cliff?



Either eqn works
 $x = 41\text{ ft}$

4. An electrician leans an extension ladder against the outside wall of a house so that it reaches an electric box 21 feet up. The ladder makes an angle of 66° with the ground. Find the length of the ladder. Round your answer to the nearest tenth of a foot if necessary.



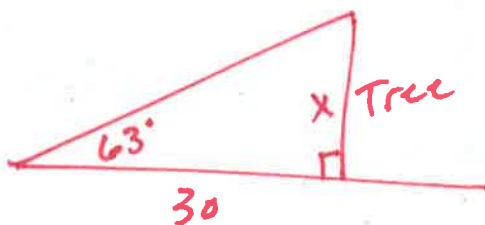
$$\sin 66 = \frac{21}{x}$$

$$x = \frac{21}{\sin 66}$$

$$x = 22.987$$

$$x = 23.0 \text{ ft}$$

5. The angle of elevation to a nearby tree from a point on the ground is measured to be 63° . How tall is the tree if the point on the ground is 30 feet from the bottom of the tree? Round your answer to the nearest tenth of a foot if necessary.



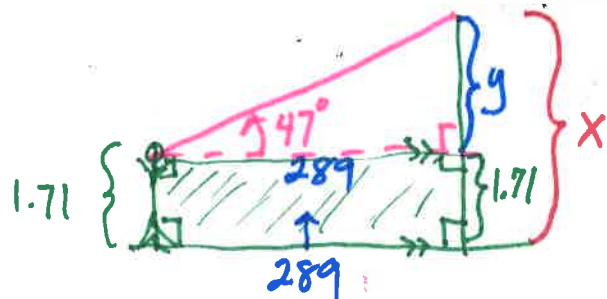
$$\tan 63 = \frac{x}{30}$$

$$x = 30 \tan 63$$

$$x = 58.9 \text{ ft}$$

you!

6. From his eye, which stands 1.71 meters above the ground, Ajay measures the angle of elevation to the top of a prominent skyscraper to be 47° . If he is standing at a horizontal distance of 289 meters from the base of the skyscraper, what is the height of the skyscraper? Round your answer to the nearest tenth of a meter if necessary.



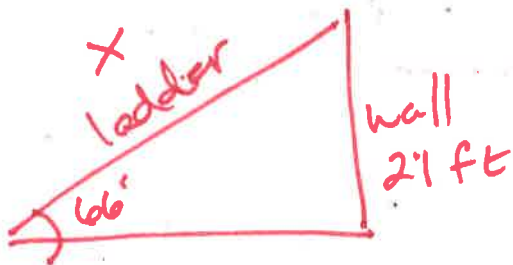
$$x = y + 1.71$$

$$\tan 47^\circ = \frac{y}{289}$$

$$y = 309.9$$

$$x = 309.9 + 1.71 = \underline{\underline{311.6 \text{ m}}}$$

4. An electrician leans an extension ladder against the outside wall of a house so that it reaches an electric box 21 feet up. The ladder makes an angle of 66° with the ground. Find the length of the ladder. Round your answer to the nearest tenth of a foot if necessary.



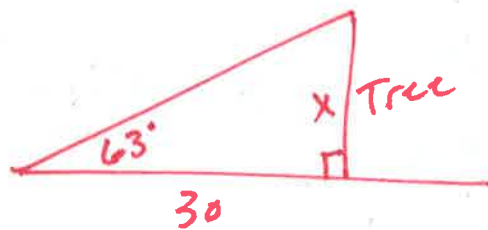
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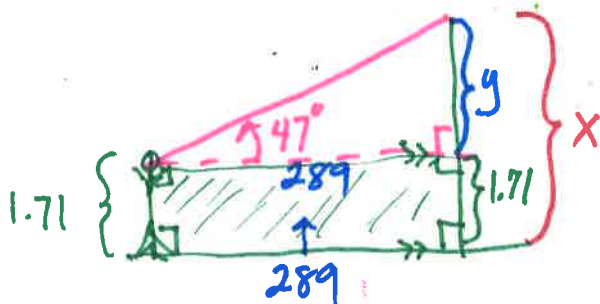
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